

Geometrical Symmetry and Tessellating Ornament

Topic: Principles of geometrical symmetry and their application in creating a sustainable, aesthetically pleasing tessellating ornament.

Integrated subjects: Mathematics, Art, Technology.

Objective

To develop students' creativity and spatial thinking by applying geometric knowledge about shapes and symmetry in the creation of surface-covering (tessellating) ornaments.

Tasks

Art

- Create an ornament following the principles of rhythm and repetition
- Arrange shapes neatly in a plane composition
- Aesthetically color and enhance the created compositions

Mathematics

- Recognize geometric shapes and their properties
- Apply types of symmetry: axial, translational, rotational, and reflective
- Understand the conditions for covering a surface without gaps or overlaps

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- **Technology**

- Create a template for a tessellating shape
- Apply construction and cutting skills when producing the model.

Tessellating Shape Definition

A tessellating shape is a shape that can completely cover a surface:

- without overlaps
- without gaps
- in such a way that the pattern can continue in any direction

A correct tessellation occurs when identical shapes fit together seamlessly, and all vertices connect neatly.

Types of Symmetry in Ornaments

- Axial symmetry
- Translational symmetry
- Rotational symmetry
- Reflective symmetry.

Practical Activity: Template Creation

1. Draw a line on thick cardboard from the upper-left corner to the upper-right corner (without cutting other edges).
2. Cut along the line and attach the parts with tape so that the edges switch places.
3. Repeat the process with another edge (top–bottom corner).
4. Use the resulting shape as a template.
5. Repeat the shape on paper without gaps or overlaps.
6. Create an ornament, color, and decorate it.

Reflection

- How did mathematics help in creating the ornament?
- Where was symmetry applied?
- How did technological solutions help to create an accurate template?
- How did aesthetics (art) contribute to the final result?

Result

- Students create a geometric, symmetry-based tessellating ornament, combining knowledge from mathematics, art, and technology into one creative project.

“Geocity – Algorithms of Sustainable Symmetry”

Methods

“Mental Laboratory”, practical MC Escher-inspired method, digital modeling

Provocation and Theoretical Foundation

Instead of standard definitions, we begin with the “Symmetry Code”:

- **Axial symmetry:** balance in parks and building facades
- **Translational symmetry:** transport lines, street rhythm, modular apartment blocks
- **Rotational symmetry:** central squares, wind farms, roundabouts
- **Tessellation:** ability to fill a plane without gaps – the most efficient use of land in a sustainable city

“M.C. Escher Challenge” – Shape Template Creation

Students create their own “Geocity” building block using a cut-and-rearrange method:

1. Start with a square or hexagon (shapes that naturally tessellate).
2. Cut a freeform shape from one side and attach it to the opposite side (translational symmetry).
3. The result is a unique shape that perfectly connects with its copies.

Mathematical task: prove that the area remains unchanged:

$$S_{\text{new}} = S_{\text{original}}$$

Digital City (ICT Lab)

Students transfer their physical design into digital space using GeoGebra, Canva, or SketchUp.

- Create a “symmetry pattern” as a future city district plan
- Use duplicate/array functions with a translation vector or rotation angle
- Model how identical modules form complex urban systems

Reflection: “City Identity”

Students present their work by answering:

- Which type of symmetry dominates your city?
- How does color influence visual dynamics even when shapes are identical?
- How does modular repetition affect the perception of urban space?

Educational Value (Alignment with Curriculum)

- Mathematics: vectors, transformations, symmetry
- Art/Architecture: spatial composition, sustainable design
- Technology: 3D modeling, digital design systems
- Soft skills: critical thinking, problem-solving, creativity

Practical Applications

Urban planning: efficient street networks and infrastructure

- Sustainability: modular construction reduces waste
- Interior design: patterned tiles, acoustic panels
- Fashion industry: textile pattern design using symmetry principles